

OSS-EXP-1

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bharath@DESKTOP-6D34C2R:~$ uname -a
Linux DESKTOP-6D34C2R 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 GNU/Linux
bharath@DESKTOP-6D34C2R:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      16
On-line CPU(s) list:         0-15
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i7-13620H
CPU family:                  6
Model:                      186
Thread(s) per core:          2
Core(s) per socket:          8
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    5836.80
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr p
ge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss
ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_
good nopl xtopology tsc_reliable nonstop_tsc cpuid
tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 p
cid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline
_timer aes xsave avx f16c rdrand hypervisor lahf_l
m abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enha
nced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjus
t bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap
clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xs
aves avx_vnni vnni umip waitpkg gfni vaes vpclmulq
dq rdpid movdiri movdir64b fsrm md_clear serialize
flush_lld arch_capabilities
Virtualization features:
Virtualization:              VT-x
Hypervisor vendor:           Microsoft
Virtualization type:         full
Caches (sum of all):
L1d:                         384 KiB (8 instances)
L1i:                         256 KiB (8 instances)
L2:                           10 MiB (8 instances)
L3:                           24 MiB (1 instance)
NUMA:
NUMA node(s):                1
NUMA node0 CPU(s):           0-15
Vulnerabilities:
Gather data sampling:         Not affected
Itlb multihit:               Not affected
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+flush_lld arch_capabilities
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  Virtualization: VT-x
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Caches (sum of all):
  L1d: 384 KiB (8 instances)
  L1i: 256 KiB (8 instances)
  L2: 10 MiB (8 instances)
  L3: 24 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-15
Vulnerabilities:
  Gather data sampling: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user po
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB condit
    ional; RSB filling; PBRBS-eIBRS SW sequence; BHI B
    HI_DIS_S
  Srbds: Not affected
  Tsx async abort: Not affected
bharath@DESKTOP-6D34C2R:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 388.6M 1 disk
sdb 8:16 0 186M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro
bharath@DESKTOP-6D34C2R:~$ ps -ef
UID PID PPID C STIME TTY TIME CMD
root 1 0 0 07:44 ? 00:00:00 /sbin/init
root 2 1 0 07:44 ? 00:00:00 /init
root 7 2 0 07:44 ? 00:00:00 plan9 --control-socket 7
root 47 1 0 07:44 ? 00:00:00 /usr/lib/systemd/systemd
systemd+ 81 1 0 07:44 ? 00:00:00 /usr/lib/systemd/systemd
root 91 1 0 07:44 ? 00:00:00 /usr/lib/systemd/systemd

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bharath 411 339 0 07:44 pts/1 00:00:00 -bash
bharath 572 338 0 07:46 pts/0 00:00:00 ps -ef
bharath@DESKTOP-6D34C2R:~$ top
top - 07:46:15 up 2 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 28 total, 1 running, 27 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.
MiB Mem : 7772.2 total, 6896.6 free, 530.0 used, 497.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7242.2 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+
586 bharath   20   0    8180    5760   3712 R   0.3   0.1   0:00.01
  1 root      20   0   24948   14704  10992 S   0.0   0.2   0:00.71
  2 root      20   0    3120    1920    1920 S   0.0   0.0   0:00.01
  7 root      20   0    3120    1792    1792 S   0.0   0.0   0:00.00
 47 root      19  -1   42180   15744  14720 S   0.0   0.2   0:00.20
 81 systemd+  20   0   22524   14080  11520 S   0.0   0.2   0:00.07
 91 root      20   0   35248   11904   8832 S   0.0   0.1   0:00.46
147 root      20   0    2880    1792    1792 S   0.0   0.0   0:00.04
152 root      20   0    4464    2816    2688 S   0.0   0.0   0:00.01
155 message+  20   0    8780    4480    4224 S   0.0   0.1   0:00.06
165 root      20   0   44076   27740  13440 S   0.0   0.3   0:00.24
172 root      20   0   18416    8832    7808 S   0.0   0.1   0:00.11
175 root      20   0  1792292  13568  11392 S   0.0   0.2   0:00.12
200 syslog    20   0   220532   4864    3968 S   0.0   0.1   0:00.07
253 _chrony    20   0   20404    5888    5248 S   0.0   0.1   0:00.03
258 root      20   0    5528    2688    2560 S   0.0   0.0   0:00.00
262 root      20   0   123440   32000   17536 S   0.0   0.4   0:00.17
264 _chrony    20   0   12076    2360    1792 S   0.0   0.0   0:00.00
289 root      20   0    5484    2560    2432 S   0.0   0.0   0:00.01
336 root      20   0    3136     768     768 S   0.0   0.0   0:00.00
337 root      20   0    3136    1156    1024 S   0.0   0.0   0:00.00
338 bharath   20   0    6260    5248    3712 S   0.0   0.1   0:00.08
339 root      20   0    8632    5248    4480 S   0.0   0.1   0:00.01
382 bharath   20   0   22320   13056   10624 S   0.0   0.2   0:00.14
388 bharath   20   0   22892    3940    2048 S   0.0   0.0   0:00.00
411 bharath   20   0    6128    5248    3712 S   0.0   0.1   0:00.03
587 root      20   0   35252    6324    3328 S   0.0   0.1   0:00.00
588 root      20   0   35252    6324    3328 S   0.0   0.1   0:00.00

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bharath@DESKTOP-6D34C2R:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.6Gi        527Mi        6.7Gi        3.5Mi        497Mi        7.1Gi
Swap:          2.0Gi          0B          2.0Gi
bharath@DESKTOP-6D34C2R:~$ mkdir OSLab
cd OSLab
bharath@DESKTOP-6D34C2R:~/OSLab$ nano exp1.c
bharath@DESKTOP-6D34C2R:~/OSLab$ gcc exp1.c -o exp1
bharath@DESKTOP-6D34C2R:~/OSLab$ ./exp1
PID = 677
bharath@DESKTOP-6D34C2R:~/OSLab$ nano exp2.c
bharath@DESKTOP-6D34C2R:~/OSLab$ gcc exp2.c -o exp2
bharath@DESKTOP-6D34C2R:~/OSLab$ ./exp2
Hello
Hello
bharath@DESKTOP-6D34C2R:~/OSLab$ nano exp3.c
bharath@DESKTOP-6D34C2R:~/OSLab$ gcc exp3.c -o exp3
bharath@DESKTOP-6D34C2R:~/OSLab$ ./exp3
I am Parent
I am Child
bharath@DESKTOP-6D34C2R:~/OSLab$ nano exp4.c
bharath@DESKTOP-6D34C2R:~/OSLab$ gcc exp4.c
bharath@DESKTOP-6D34C2R:~/OSLab$ ./a.out
Parent PID = 884
Child PID = 885
Child PID = 885
Parent PID = 884
bharath@DESKTOP-6D34C2R:~/OSLab$ nano exp5.c
bharath@DESKTOP-6D34C2R:~/OSLab$ gcc exp5.c
bharath@DESKTOP-6D34C2R:~/OSLab$ ./a.out
Child Executing
Parent Executing
bharath@DESKTOP-6D34C2R:~/OSLab$ nano exp6.c
bharath@DESKTOP-6D34C2R:~/OSLab$ gcc exp6.c -o exp6
bharath@DESKTOP-6D34C2R:~/OSLab$ ./exp6
PID = 1001
PID = 1000
PID = 1003
PID = 1002
bharath@DESKTOP-6D34C2R:~/OSLab$ nano exp7.c
bharath@DESKTOP-6D34C2R:~/OSLab$ gcc exp7.c -o exp7
bharath@DESKTOP-6D34C2R:~/OSLab$ ./exp7
Parent Waiting...
Child executing ls...
bharath@DESKTOP-6D34C2R:~/OSLab$ total 124
-rwxr-xr-x 1 bharath bharath 16032 Aug  2 07:51 a.out
bharath@DESKTOP-6D34C2R:~/OSLab$ nano exp8.c
bharath@DESKTOP-6D34C2R:~/OSLab$ pwd
/home/bharath/OSLab
bharath@DESKTOP-6D34C2R:~/OSLab$ cd OSLAB
-bash: cd: OSLAB: No such file or directory
bharath@DESKTOP-6D34C2R:~/OSLab$ cd OSLab
-bash: cd: OSLab: No such file or directory
bharath@DESKTOP-6D34C2R:~/OSLab$ cd
bharath@DESKTOP-6D34C2R:~$ pwd
/home/bharath
bharath@DESKTOP-6D34C2R:~$ cd OSLab
bharath@DESKTOP-6D34C2R:~/OSLab$ ls
a.out  exp1.c  exp2.c  exp3.c  exp5.c  exp6.c  exp7.c
exp1   exp2    exp3    exp4.c  exp6    exp7
bharath@DESKTOP-6D34C2R:~/OSLab$ ls exp7
exp7
bharath@DESKTOP-6D34C2R:~/OSLab$ cd exp7
-bash: cd: exp7: Not a directory
bharath@DESKTOP-6D34C2R:~/OSLab$ ls
a.out  exp1.c  exp2.c  exp3.c  exp5.c  exp6.c  exp7.c
exp1   exp2    exp3    exp4.c  exp6    exp7
bharath@DESKTOP-6D34C2R:~/OSLab$ ls -l exp7
-rwxr-xr-x 1 bharath bharath 16032 Aug  2 07:52 exp7
bharath@DESKTOP-6D34C2R:~/OSLab$ cat exp4.c
#include <stdio.h>

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